

Inverter MODBUS Communication Protocol

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Revision History

Version	Date	Reason / Description
V0.1	2018/10	Initial document creation

1. Introduction

This document describes the Modbus communication protocol for Foshan Jin Guangyuan Power Technology Co., Ltd. inverters, applicable to inverter communications. This protocol references *GBT19582-2008 Industrial Automation Network Specification Based on Modbus Protocol*.

2. MODBUS Protocol Description

2.1 Modbus Address Rules

Modbus operates on a master-slave communication architecture. Communication is initiated by the master device, and the corresponding slave device responds. The master has no address; slave addresses range from **1 to 247**, and **0** is the broadcast address. Slave addresses are unique on a Modbus serial bus.

2.2 Modbus Protocol Frame

The Modbus protocol frame consists of an Address Field, Function Code, Data Field, and Checksum Field.

Address Field	Function Code	Data Field	Checksum Field
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Table 2.1.1 General Modbus Frame

2.3 RTU Transmission Mode

2.3.1 Byte Format

Each byte contains: 1 start bit, 8 data bits (least significant bit sent first), no parity bit, 1 stop bit, totaling 10 bits. Each character or byte is transmitted in this order (left to right): LSB ... MSB.

2.3.2 RTU Message Frames

An RTU message frame contains: Slave Address, Function Code, Data Field, and CRC Checksum. The maximum RTU message frame length is 250 bytes, where the Data Field length is up to 246 bytes.

Format	Slave Address	Function Code	Data	CRC Checksum
Byte Count	1 Byte	1 Byte	0 ~ 252 Bytes	2 Bytes (Low Byte, High Byte)

Table 2.3.2a RTU Message Frame Structure

In RTU mode, an idle interval of at least 3.5 character times separates message frames. The entire message frame must be transmitted as a continuous stream of characters.

If an idle interval greater than 1.5 character times occurs between two characters, the message frame is considered incomplete, and the receiving station must discard it.

2.3.3 CRC Checksum

The CRC field is a 16-bit value composed of two 8-bit bytes, appended to the message as the final field (low-order byte first, followed by high-order byte). All register data (*INT16*) uses swapped high-low byte transmission order, including message headers.

2.4 Exception Codes

Code	Description	Remarks
01H	Illegal Function Code	The function code received in the query is an unallowable operation.
02H	Illegal Data Address	The data address received in the query is an unallowable address.
03H	Illegal Data Value	The value included in the query is an unallowable value.
06H	Slave Device Busy	The slave is processing a long-duration command.

Table 2.4.1 Exception Code Details

2.5 Storage Area Address Range

This protocol restricts storage area address ranges into groups according to the rules in Table 2.5.1.

Name	Type	Read / Write	Storage Unit Address	Function Code
Output Registers	Word	Write-Only	200 - 233	06H, 10H
Input Registers	Word	Read-Only	000 - XXX	04H
Holding Registers	Word	Read/Write	000 - 199	03H, 06H, 10H

Table 2.5.1 Storage Area Address Rules

2.6 Function Code List

Function Code Index	Function Code	Description	Remarks
1	01H	Read Output Status (Coils)	
2	02H	Read Input Status (Discrete Inputs)	
3	03H	Read Holding Registers	
4	04H	Read Input Registers	
5	05H	Write Single Output Status	
6	06H	Write Single Register	
7	0FH	Write Multiple Output Status	
8	10H	Write Multiple Registers	
9	14H	Read File Record	
10	2BH	Read Device Identification	

Table 2.5.2 Function Code List

3. Protocol Application

The device uses RS232 and RS485 as physical layer interfaces. Data signal transmission rate (Baud rate): **9600**; Communication data format: **8, N, 1**; ID: **01**.

Function Code Application Mapping Table

Application Content	Type	Function Code	Operation
Control Status	Bit	01H, 05H	Read, Write
Control Command / Status / Analogs	Word, Bit, Word	06H, 02H, 04H	Write-Only, Read-Only, Read
Software Version	-	2BH	Read
History Records	-	14H	Read
User Settings	Word	03H, 06H	Read/Write
System Date & Time	Word	03H, 10H	Read/Write
System Self-Check Cycle	Word	03H, 10H	Read/Write
History Record Count	Word	03H, 10H	Read/Write

Table 3.0.1 Function Code Application Table

3.1 Input Register Data Content (Whole Machine Data)

Address	Content	Size	Unit & Range
000	Input AC Voltage	2Byte	0 ~ 5000 (0.1V)
001	Input Frequency	2Byte	0 ~ 5000 (0.1Hz)
002	Output Voltage	2Byte	0 ~ 5000 (0.1V)
003	Output Frequency	2Byte	0 ~ 5000 (0.1Hz)
004	Output Current	2Byte	0 ~ 20000 (0.1A)
005	Reserved	2Byte	-
006	Output Load Rate	2Byte	0 ~ 300 (%)
007	Battery Voltage	2Byte	0 ~ 5000 (0.1V)
008 / 009	Reserved / Battery Capacity Rate	2Byte / 2Byte	0 ~ 100 (%)
012	DC Bus Current	2Byte	0.1A (0 ~ 65535)
013	Internal Temperature	2Byte	0 ~ 2000 (0.1 °C)
014	Ambient Temperature	2Byte	0 ~ 2000 (0.1 °C)
031	Setting Status Bits	2Byte	-
032	Running Status Bits	2Byte	-
033	Warning Status Bits	2Byte	-
034	Fault Status Bits	2Byte	-
035	Event Code	2Byte	-

Table 3.1.1 Whole Machine Data Register Map